

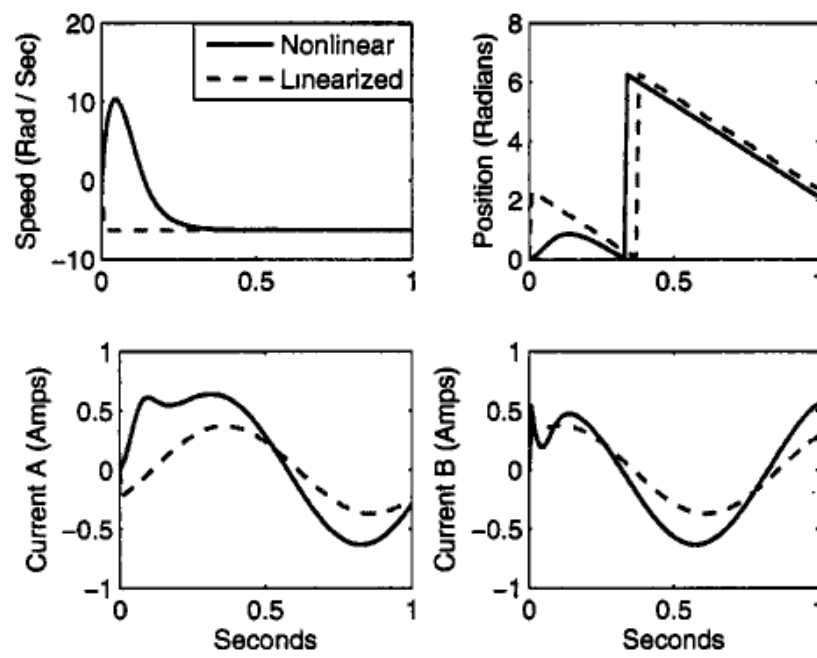


Figure 1.1 Example 1.4 comparison of nonlinear and linearized motor simulations.

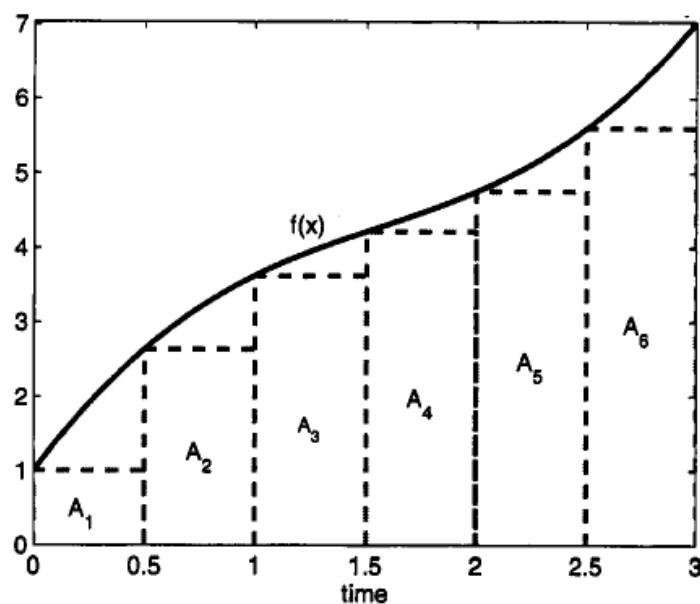


Figure 1.2 An illustration of rectangular integration. We have $\dot{x} = f(x)$, so $x(t)$ is the area under the $f(x)$ curve. This area can be approximated as the sum of the rectangular areas A_i . That is, $x(0.5) \approx A_1$, $x(1) \approx A_1 + A_2$, $\dots$.

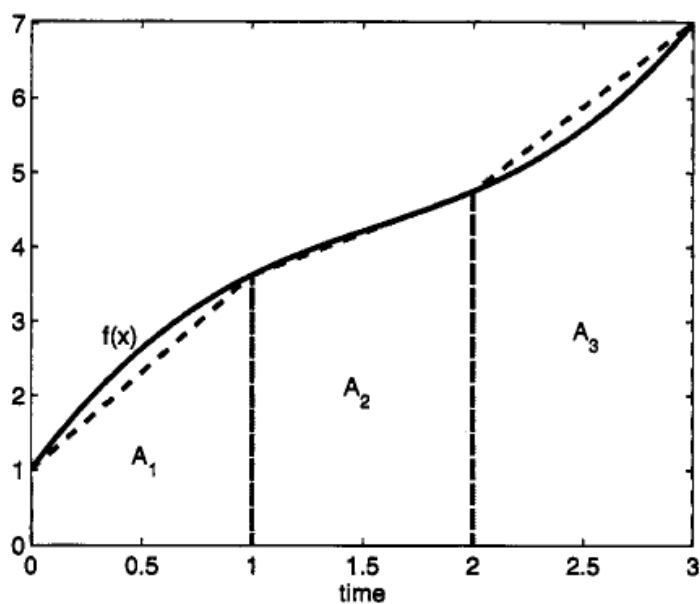


Figure 1.3 An illustration of trapezoidal integration. We have $\dot{x} = f(x)$, so $x(t)$ is the area under the $f(x)$ curve. This area can be approximated as the sum of trapezoidal areas A_i . That is, $x(1) \approx A_1$, $x(2) \approx A_1 + A_2$, and $x(3) \approx A_1 + A_2 + A_3$.

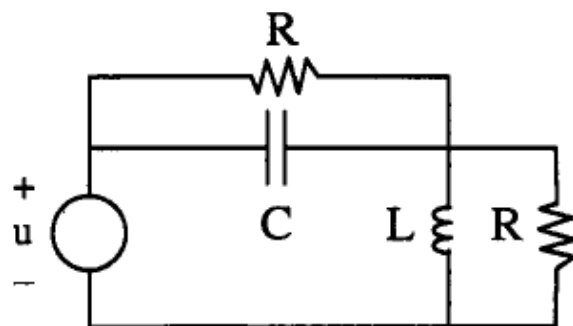


Figure 1.4 RLC circuit for Example 1.8.

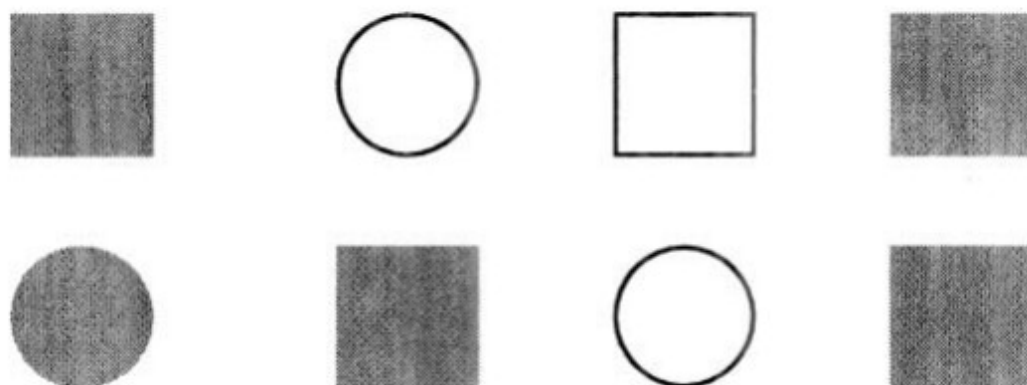


Figure 2.1 Some shapes for illustrating probability and Bayes' Rule.

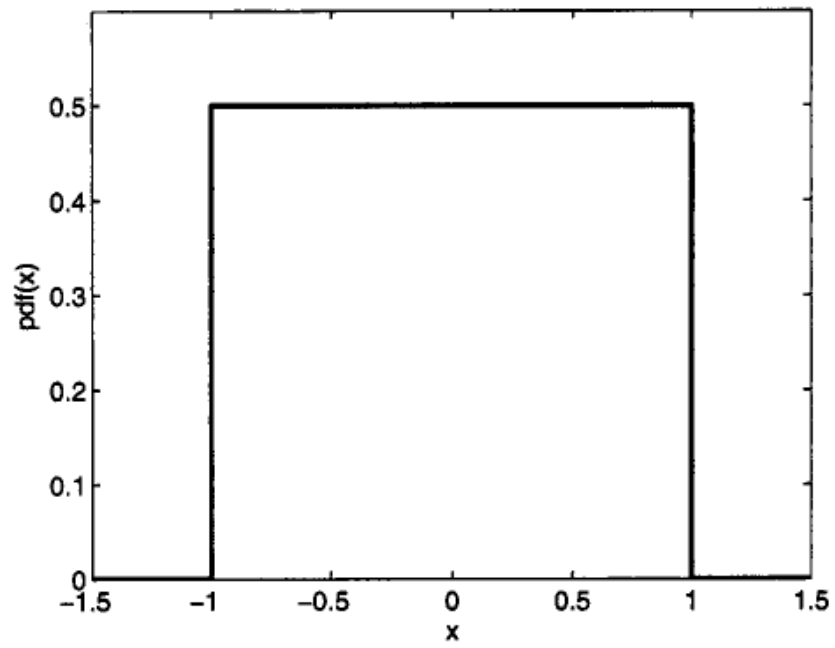


Figure 2.2 Probability density function of an RV uniformly distributed between ± 1 .

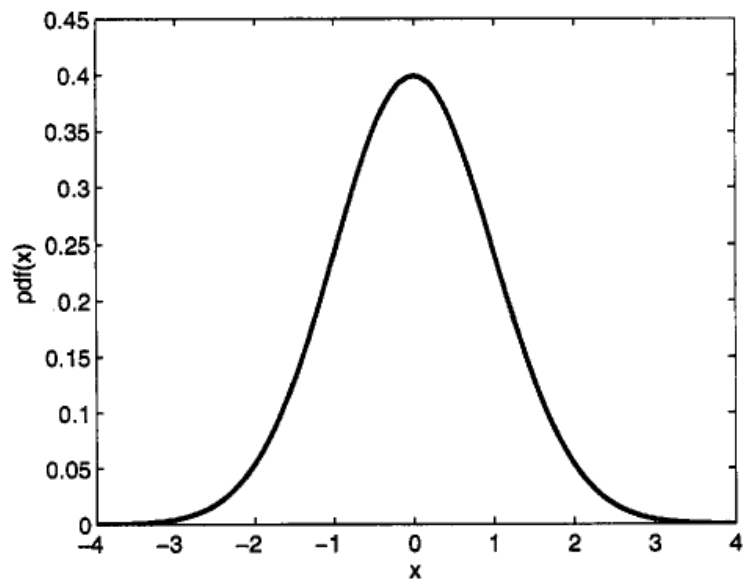


Figure 2.3 Probability density function of a Gaussian RV with a mean of zero and a variance of one.

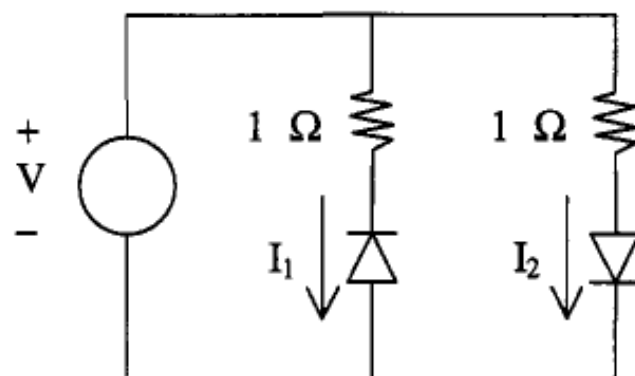


Figure 2.4 Circuit for Example 2.11.

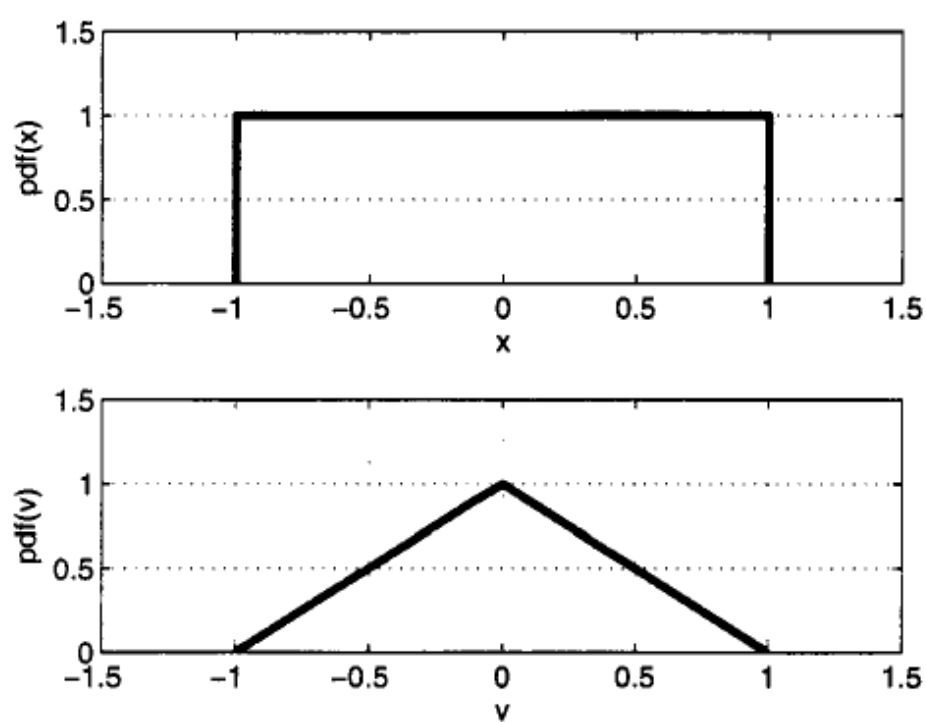


Figure 2.5 pdf's for Problem 2.13 [Sch73].

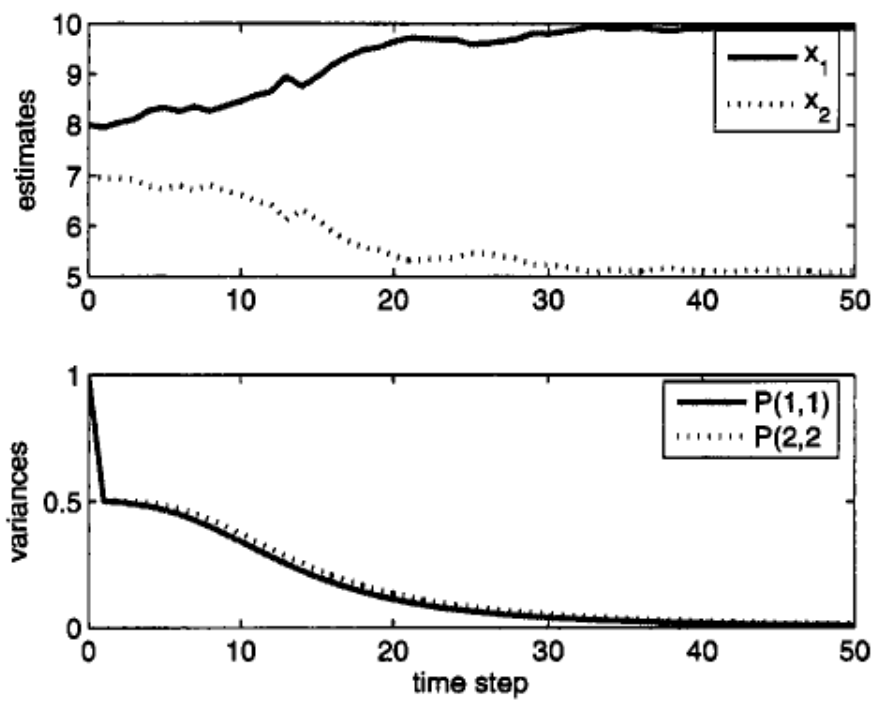


Figure 3.1 Parameter estimates and estimation variances for Example 3.5.

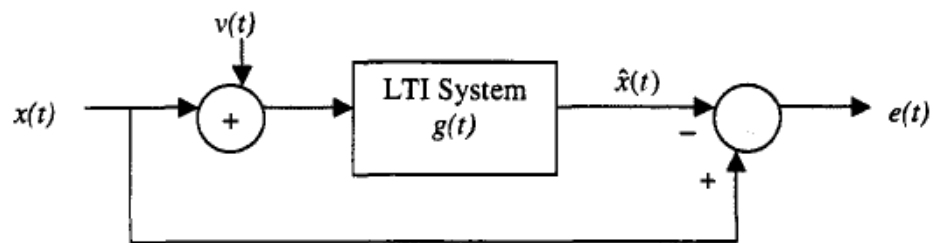


Figure 3.2 Wiener filter representation.

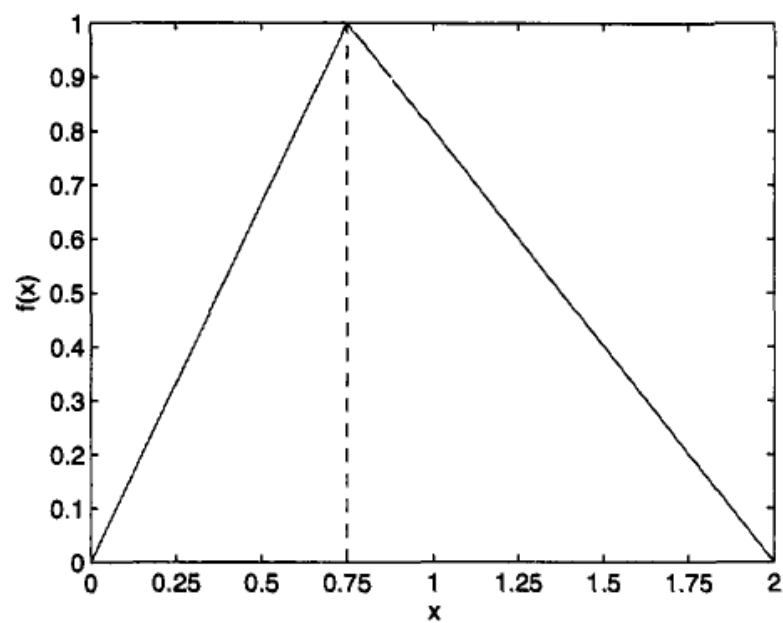


Figure 3.3 pdf for Problem 3.8.

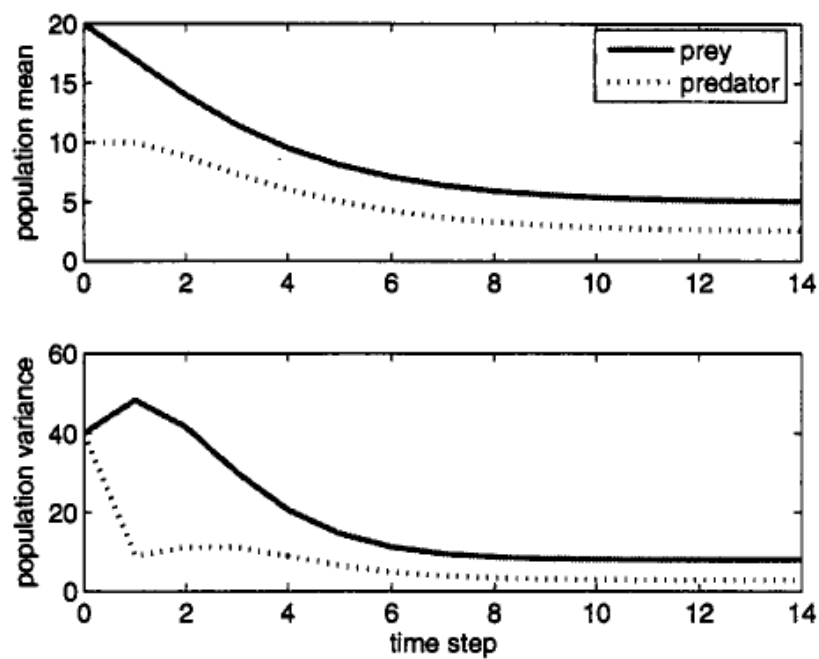


Figure 4.1 State means and variances for Example 4.1.

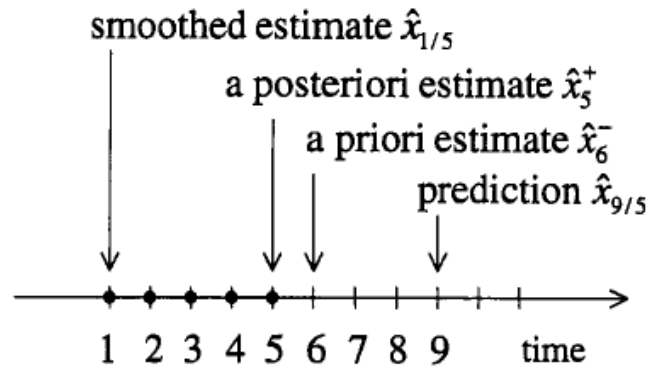


Figure 5.1 Time line showing the relationship between the *a posteriori*, *a priori*, smoothed, and predicted state estimates. In this figure, we suppose that we have received measurements at times up to and including $k = 5$. An estimate of the state at $k < 5$ is called a smoothed estimate. An estimate of the state at $k = 5$ is called the *a posteriori* estimate. An estimate of the state at $k = 6$ is called the *a priori* estimate. An estimate of the state at $k > 6$ is called the prediction.

These relationships are depicted in Figure 5.2. The figure shows that after we process the measurement at time $(k-1)$, we have an estimate of x_{k-1} (denoted $\hat{x}_{k-1}^+$) and the covariance of that estimate (denoted P_{k-1}^+). When time k arrives, before we process the measurement at time k we compute an estimate of x_k (denoted $\hat{x}_k^-$) and the covariance of that estimate (denoted P_k^-). Then we process the measurement at time k to refine our estimate of x_k . The resulting estimate of x_k is denoted $\hat{x}_k^+$, and its covariance is denoted P_k^+ .

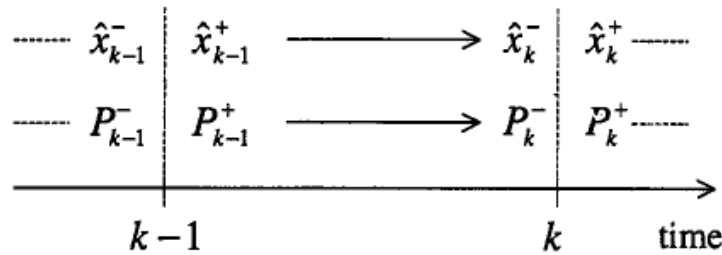


Figure 5.2 Timeline showing *a priori* and *a posteriori* state estimates and estimation-error covariances.

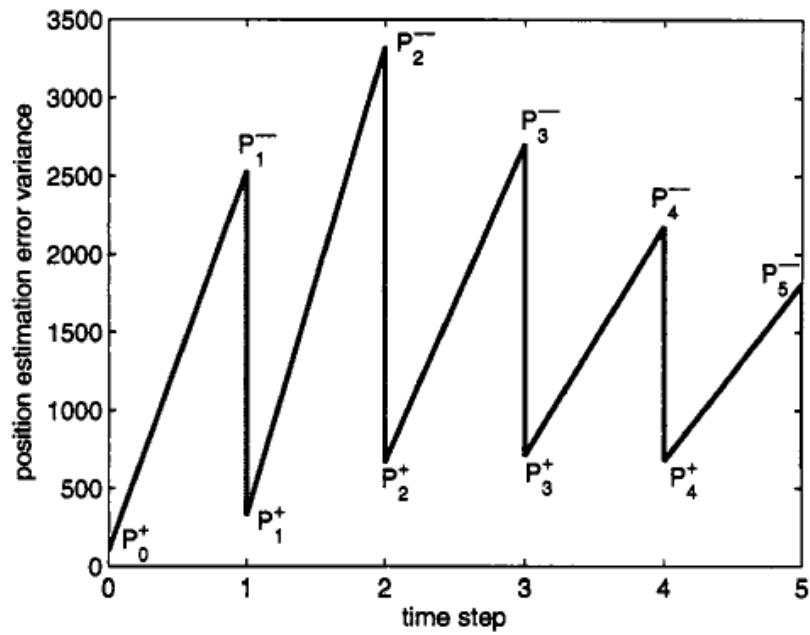


Figure 5.3 The first five time steps of the *a priori* and *a posteriori* position-estimation-error variances for Example 5.1.

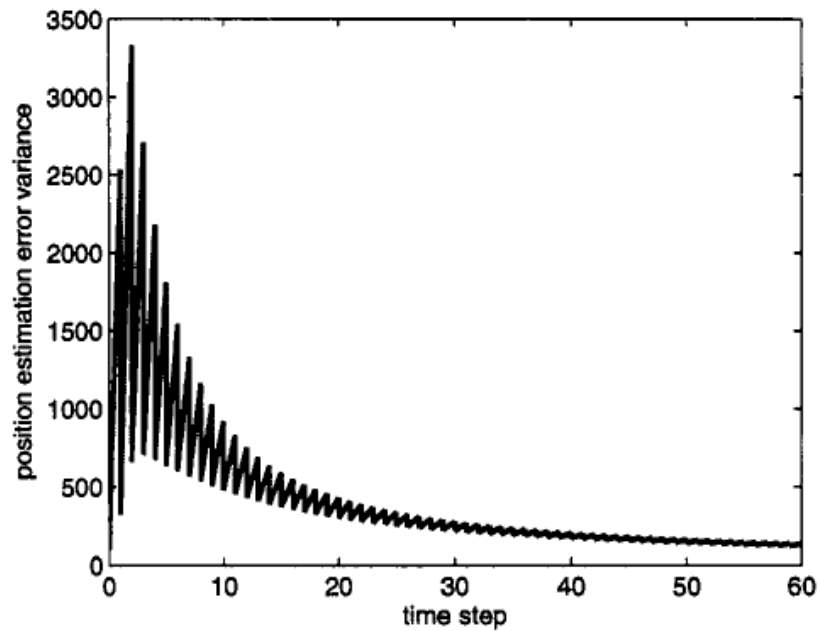


Figure 5.4 The first 60 time steps of the *a priori* and *a posteriori* position-estimation-error variances for Example 5.1.

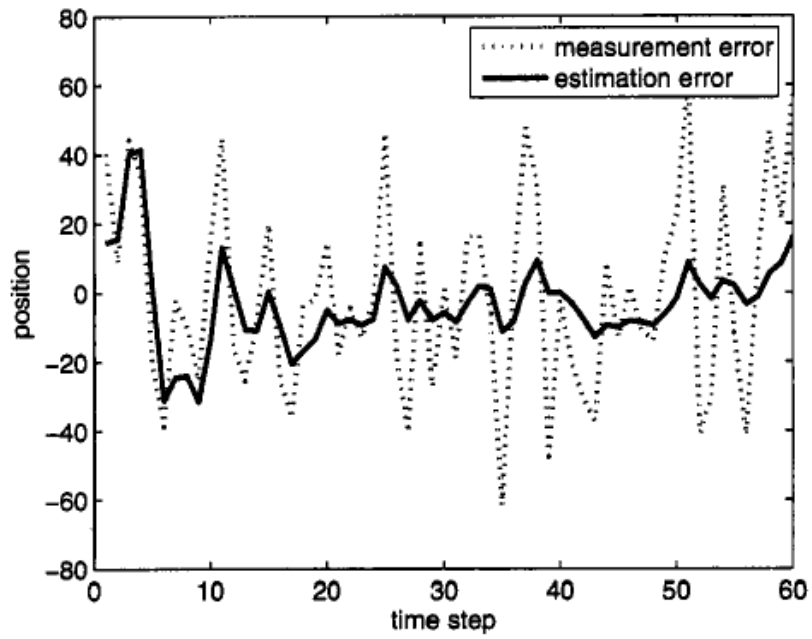


Figure 5.5 The position-measurement error and position estimation error for Example 5.1.

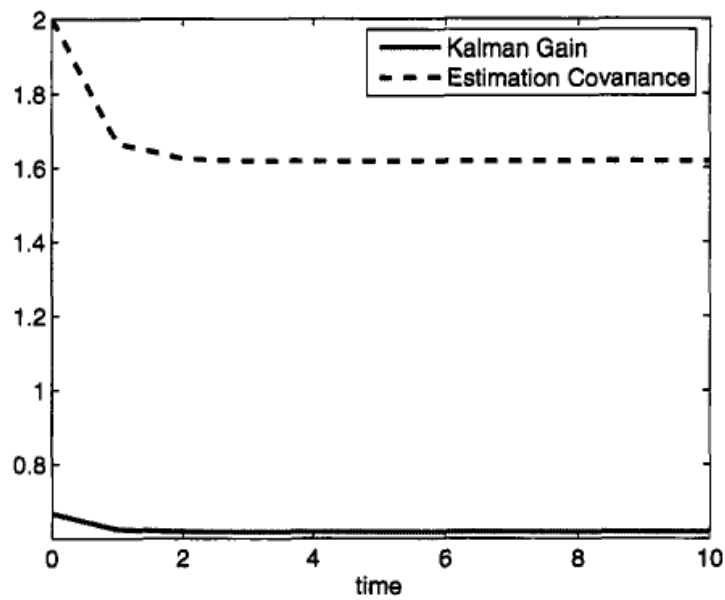


Figure 5.6 Estimation covariance and Kalman gain as a function of time for Example 5.2. The covariance and gain converge to steady-state values.

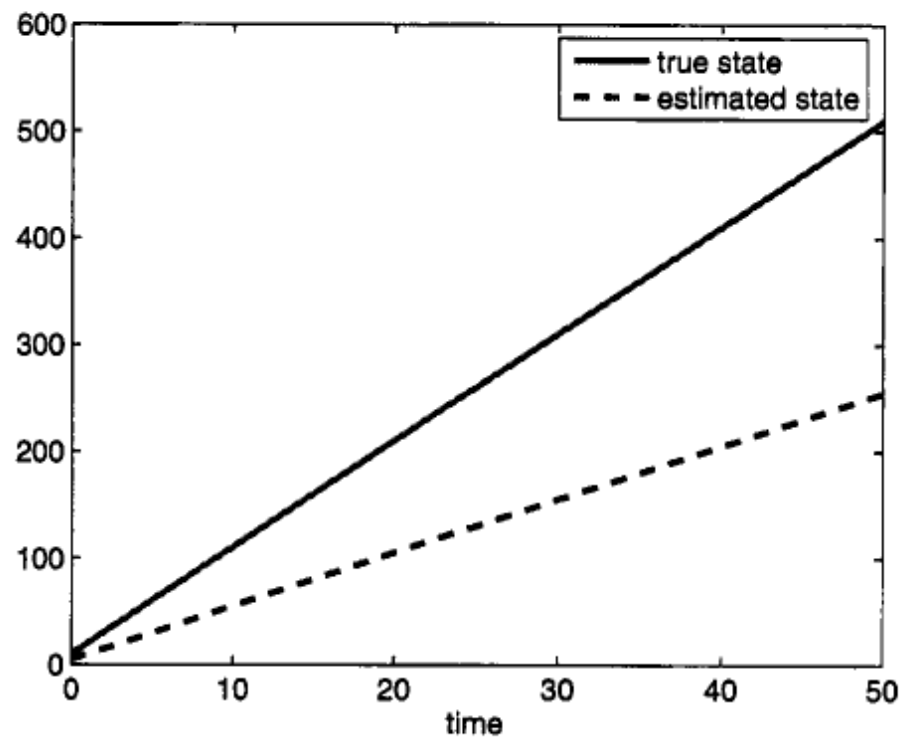


Figure 5.7 Kalman filter divergence due to mismatching.

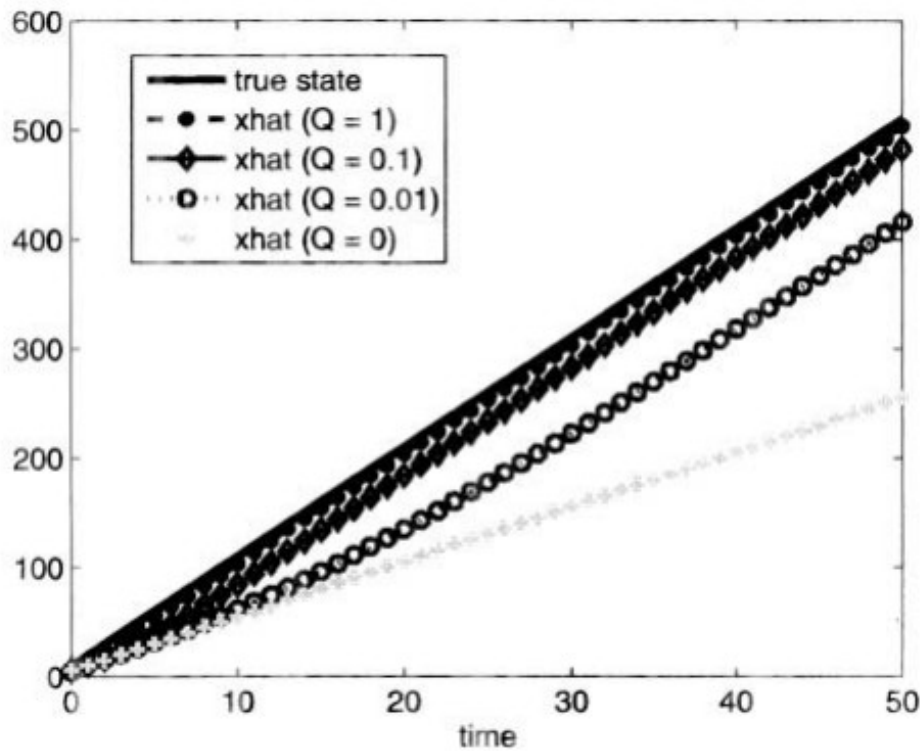
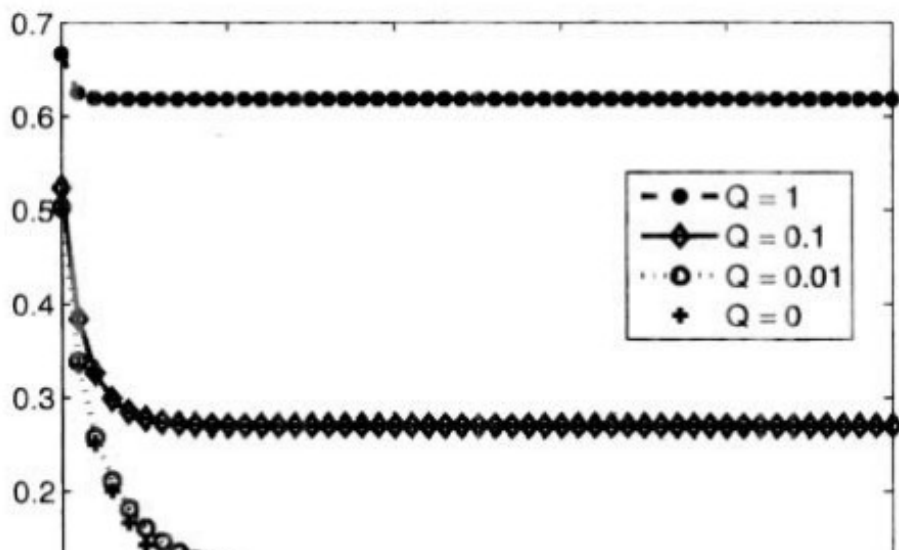


Figure 5.8 Kalman filter improvement due to fictitious process noise.

measurements [see the $\hat{x}_k^+$ expression in Equation (5.61)]. This compensates for modeling errors. As shown later in Section 7.4, the fading-memory filter accomplishes the same thing in a different way. Also note from Figure 5.9 that the steady-state Kalman gain is approximately 0.62 when $Q = 1$. This matches the results of Example 5.2.



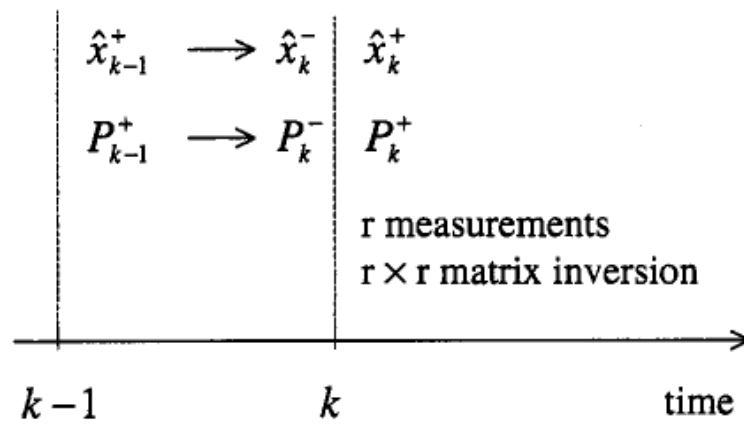


Figure 6.1 The measurement-update equation of the standard Kalman filter requires an $r \times r$ matrix inversion, where r is the number of measurements.

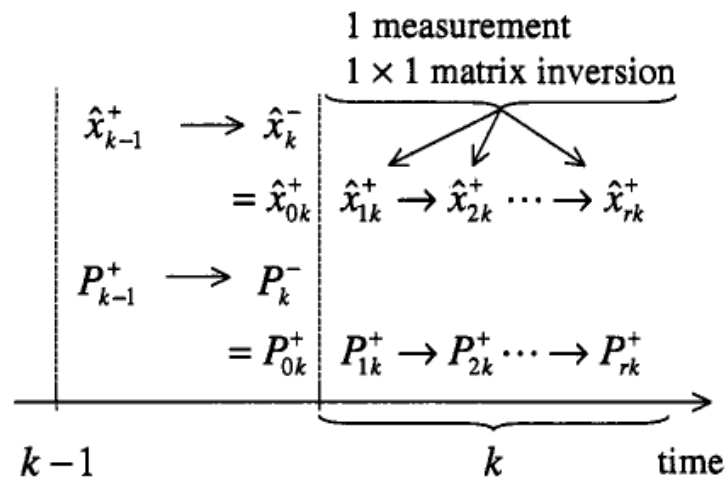


Figure 6.2 The measurement update equation of the sequential Kalman filter requires r scalar divisions (where r is the number of measurements) because the measurements at each time step are processed sequentially. This is in contrast to the standard Kalman filter processing that is depicted in Figure 6.1.

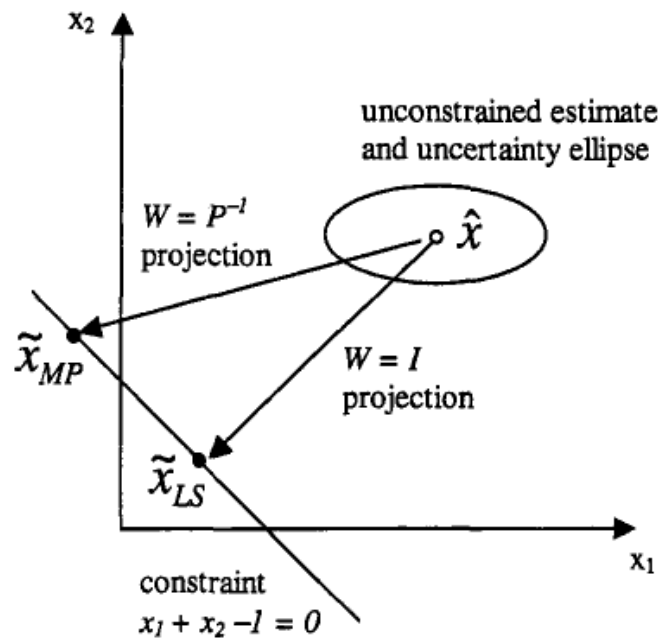


Figure 7.1 In Example 7.11, the unconstrained estimate violates the equality constraint. The least squares approach to constrained estimation projects the estimate in the direction perpendicular to the constraint surface. The maximum probability approach projects the estimate in the direction P^{-1} relative to the constraint surface.

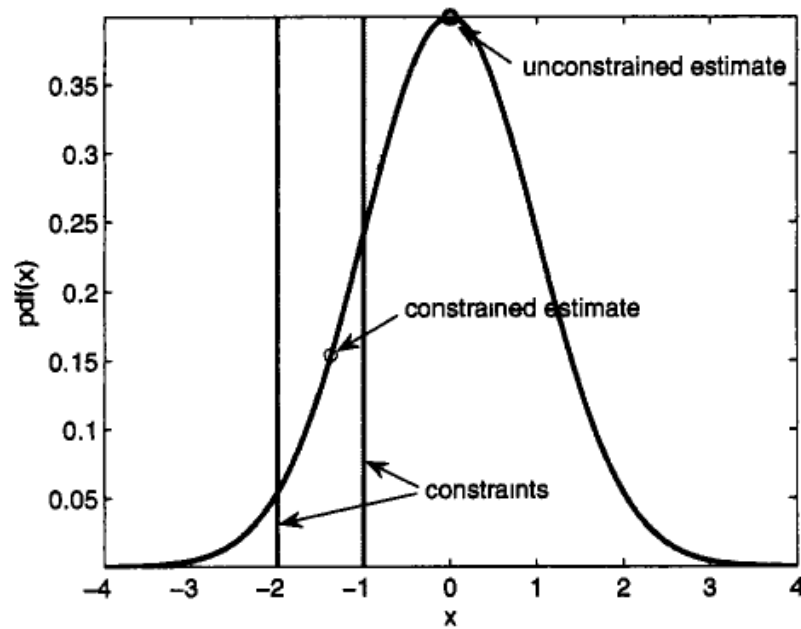


Figure 7.2 The unconstrained estimate at $x = 0$ violates the constraints. The constrained estimate, which is at $x \approx -1.38$, is the centroid of the truncated pdf.

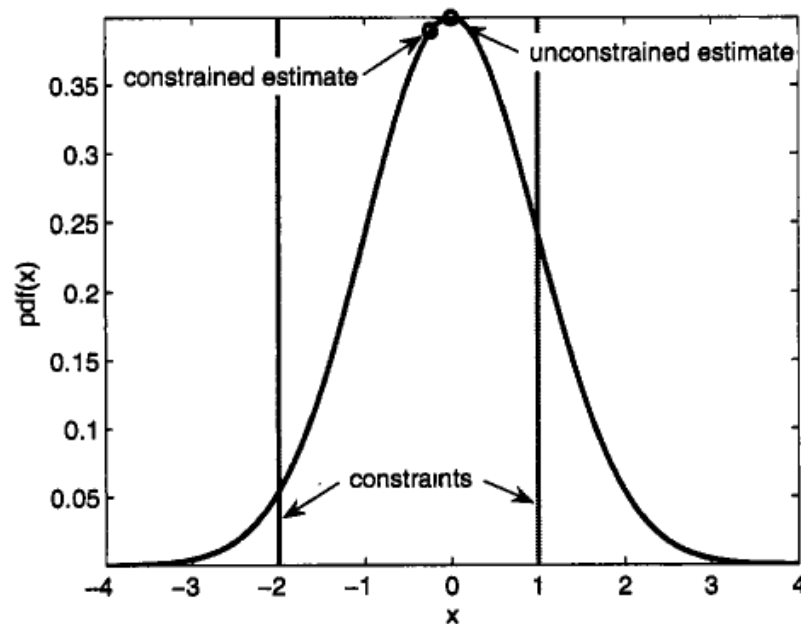


Figure 7.3 The unconstrained estimate at $x = 0$ satisfies the constraints. Nevertheless, the truncation approach to constrained estimation shifts the estimate to the centroid of the truncated pdf, which is at $x \approx -0.23$.

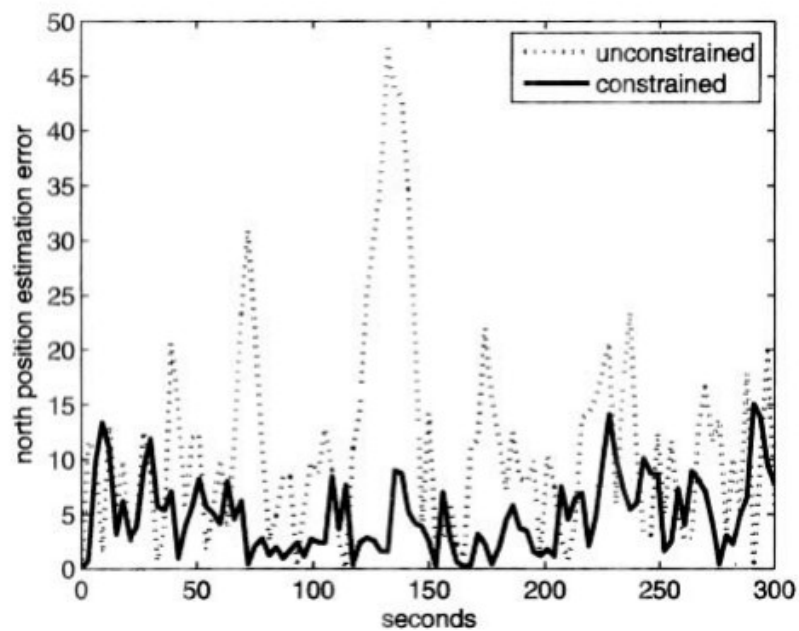


Figure 7.4 North position estimation error magnitude of the unconstrained and constrained Kalman filters for Example 7.12.

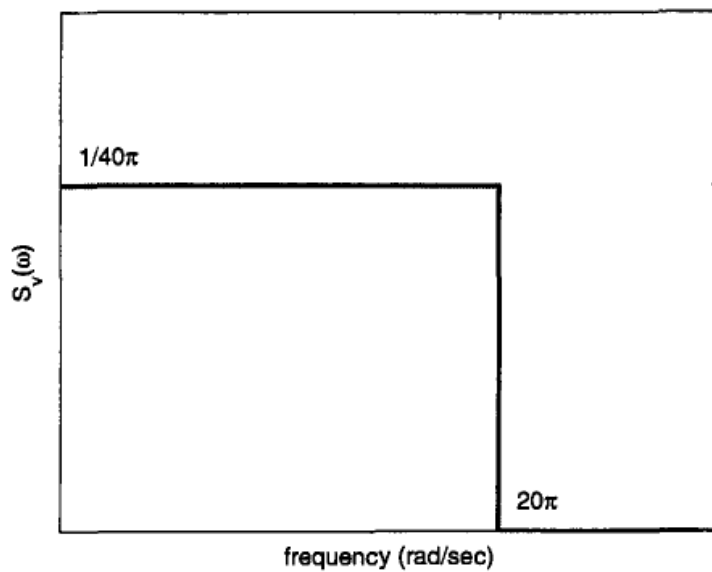


Figure 8.1 Power spectrum of bandlimited measurement noise for Example 8.4.

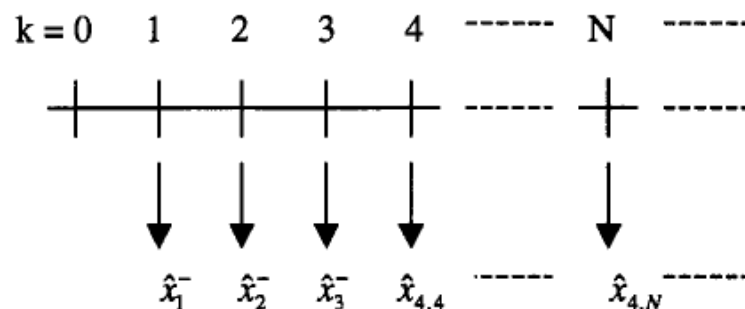


Figure 9.1 Fixed-point smoothing. We desire an estimate of x_4 . Up until $k = 4$, the standard Kalman filter operates. At $k = 4$, we have $\hat{x}_4^- = \hat{x}_{4,4}$, which is the estimate of x_4 based on measurements up to and including y_3 . As time progresses, we continue to refine our estimate of x_4 based on an increasing number of measurements. At time $k = N$, we have $\hat{x}_{4,N}$, which is the estimate of x_4 based on measurements up to and including time $N - 1$.

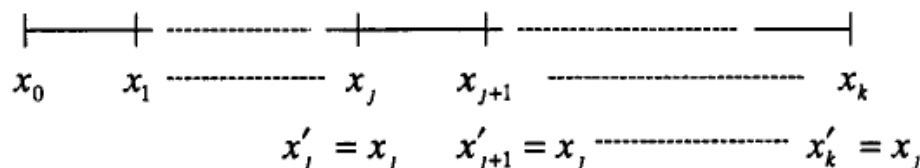


Figure 9.4 This illustrates the idea that is used to obtain the fixed-point smoother. A fictitious state variable x' is initialized as $x'_j = x_j$ and from that point on has an identity state transition matrix. The *a priori* estimate of x'_k is then equal to $\hat{x}_{j,k}$.

Measurement noise standard deviation	Percent Improvement
0.1	99.7
1	96.6
10	59.3
100	13.7
1000	0.2

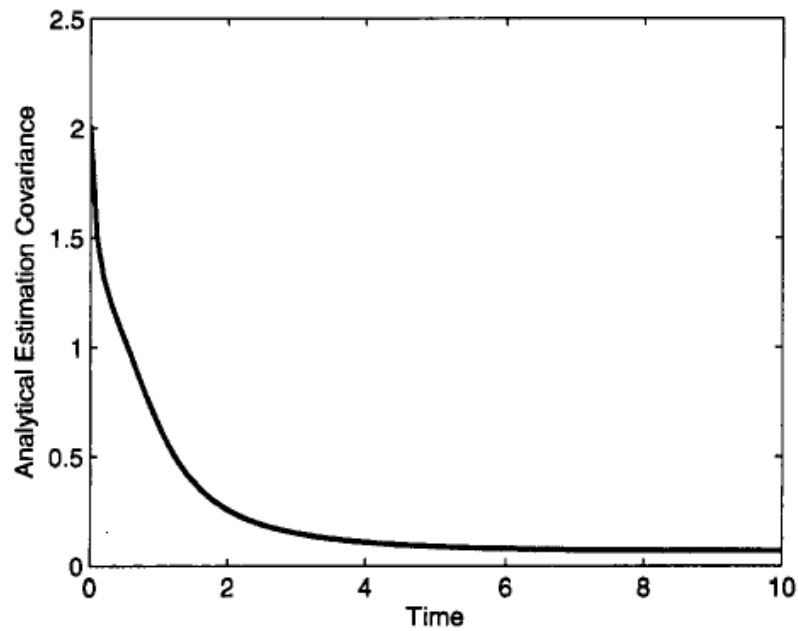


Figure 9.5 This shows the trace of the estimation-error covariance of the smoothed estimate of the state at the first time step for Example 9.1. As time progresses and we process more measurements, the covariance decreases, eventually reaching steady state.

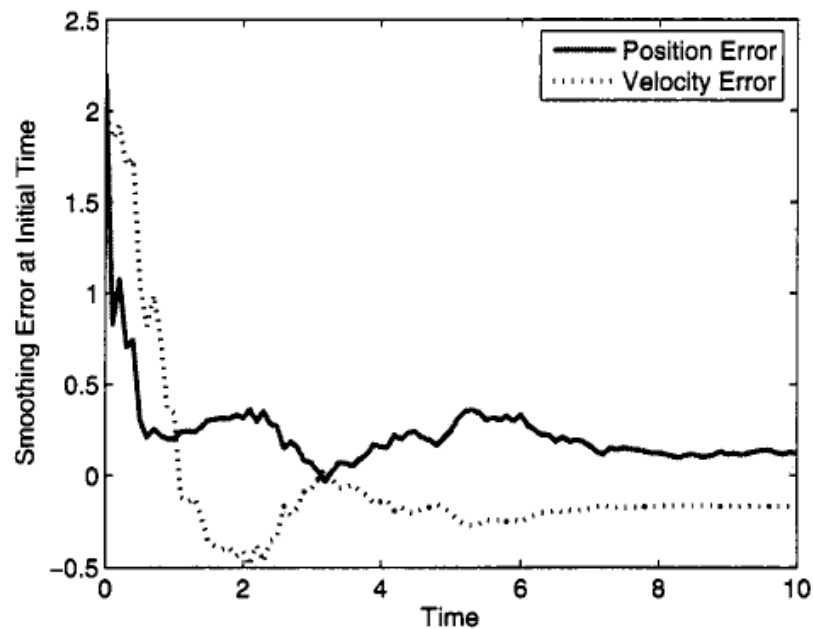


Figure 9.6 This shows typical estimation errors of the smoothed estimate of the state at the first time step for Example 9.1. As time progresses and we process more measurements, the estimation error decreases, and its standard deviation eventually reaches steady state.

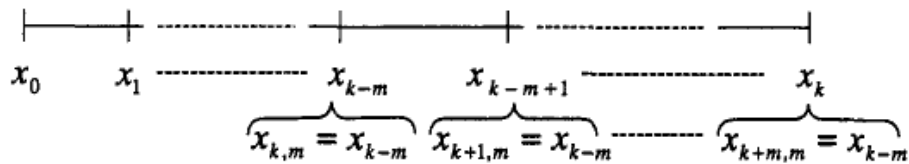


Figure 9.7 This illustrates the idea that is used to obtain the fixed-lag smoother. A fictitious state variable $x_{k,m}$ is initialized as $x_{k,m} = x_{k-m}$ and from that point on has an identity state transition matrix. The *a posteriori* estimate of $x_{k+m,m}$ is then equal to $\hat{x}_{k-m,k}$.

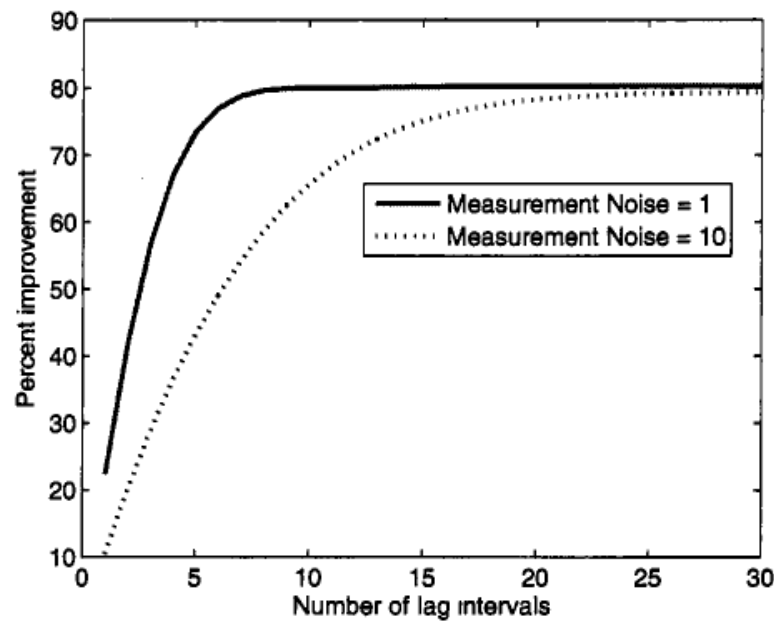


Figure 9.8 This shows the percent improvement of the trace of the estimation-error covariance of the smoothed estimate of the state (relative to the standard Kalman filter) for Example 9.2. As the number of lag intervals increases, the estimation error of the smoother decreases and the percent improvement increases. Also, as the measurement noise decreases, the improvement due to smoothing is more dramatic.

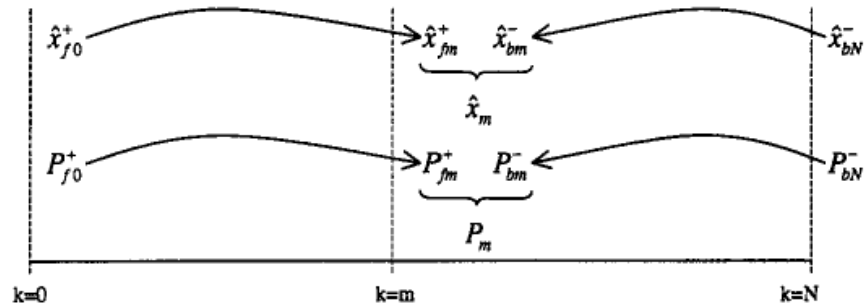


Figure 9.9 This figure illustrates the concept of the forward-backward smoother. The forward filter is run to obtain *a posteriori* estimates and covariances up to time m . Then the backward filter is run to obtain *a priori* estimates and covariances back to time m (i.e., *a priori* from a reversed time perspective). Then the forward and backward estimates and covariances at time m are combined to obtain the final estimate $\hat{x}_m$ and covariance P_m .

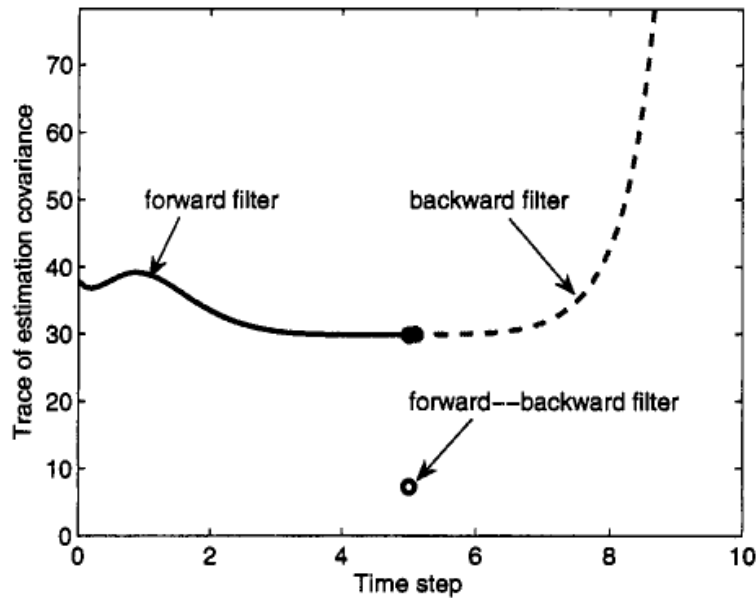


Figure 9.10 This shows the trace of the estimation-error covariance for Example 9.3. The forward filter runs from $t = 0$ to $t = 5$, the backward filter runs from $t = 10$ to $t = 5$, and the trace of the covariance of the smoothed estimate is shown at $t = 5$.

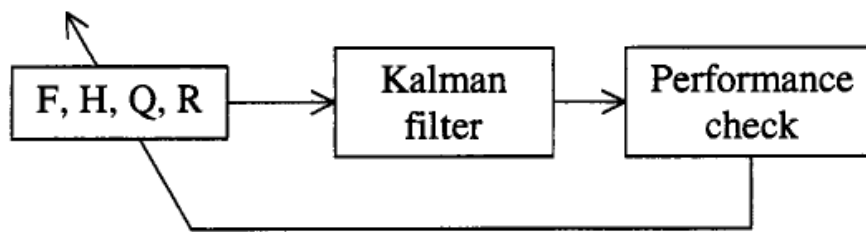


Figure 10.1 This figure illustrates how the performance of a Kalman filter can be used to tune the values of F , H , Q , and R in order to obtain residual statistics that agree with theory. Alternatively, the Kalman gain K could be tuned directly.

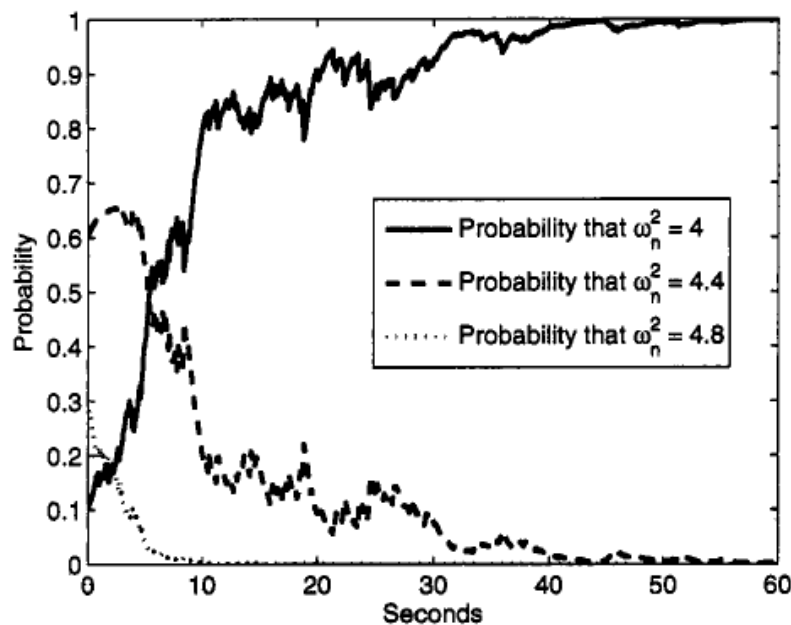


Figure 10.2 Parameter probabilities for the multiple-model Kalman filter for Example 10.1. The true parameter value is 4, and the filter converges to the correct parameter after a few seconds.

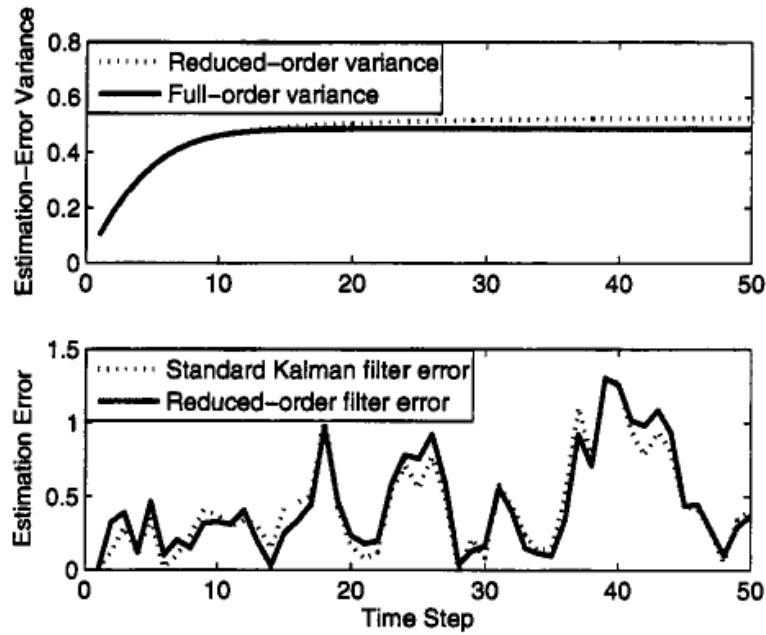


Figure 10.3 Results for Example 10.2. The top figure shows the analytical estimation-error variances for the first state for the full-order filter and the reduced-order filter. As expected, the reduced-order filter has a higher estimation-error variance, but the small degradation in performance may be worth the computational savings, depending on the application. The bottom figure shows typical error magnitudes for the estimate of the first state for the full-order filter and the reduced-order filter. The reduced-order filter has slightly larger estimation errors.

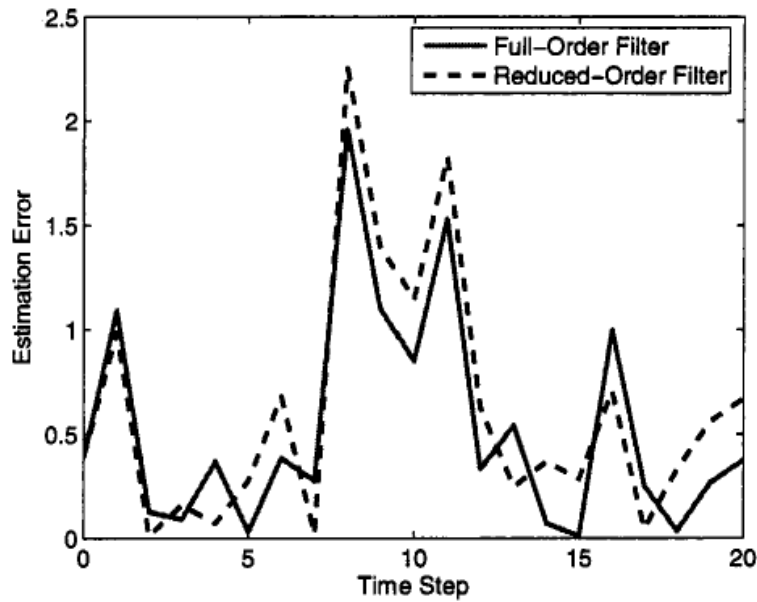


Figure 10.4 Results for Example 10.3. Typical error magnitudes for the estimate of the first state for the full-order filter and the reduced-order filter. The reduced-order filter has only slightly larger estimation errors.

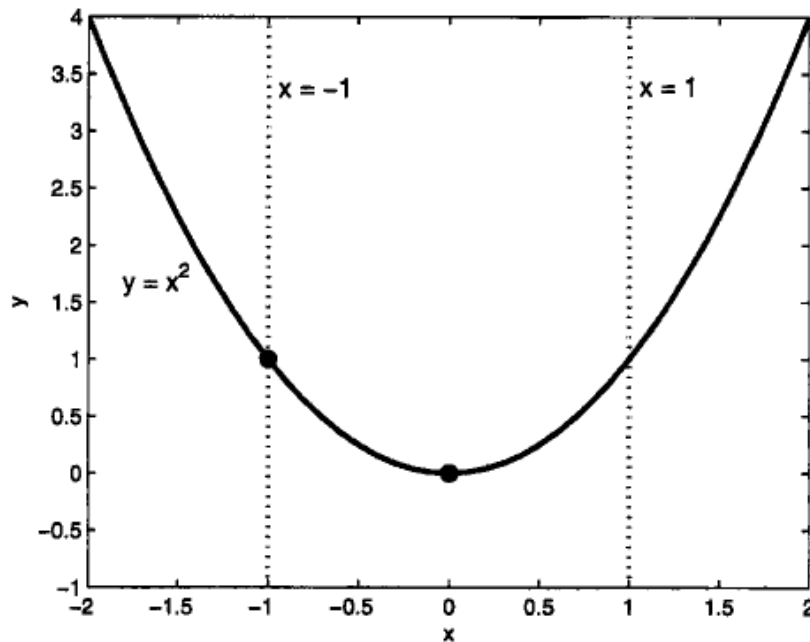


Figure 11.1 This illustrates the constrained minimization of x^2 . If the constraint is $x - 1 \leq 0$, then the constrained minimum is equal to the unconstrained minimum and occurs at $x = 0$. If the constraint is $x + 1 \leq 0$, then the constrained minimum can be solved by enforcing the equality constraint $x + 1 = 0$ and occurs at $x = -1$.

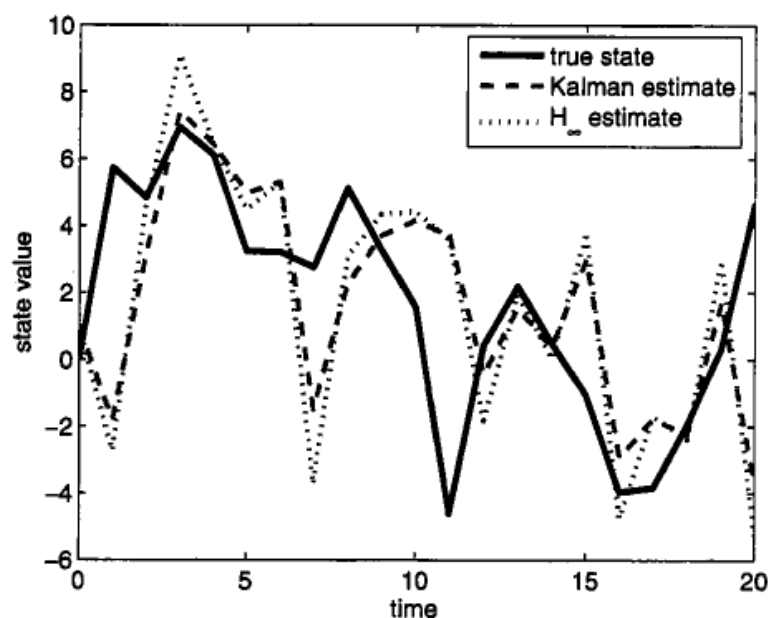


Figure 11.2 Example 11.2 results. Kalman and H_∞ filter performance when the noise statistics are known. The Kalman gain is 0.62 and the H_∞ gain is 0.82. The Kalman filter performs about 12% better than the H_∞ filter.

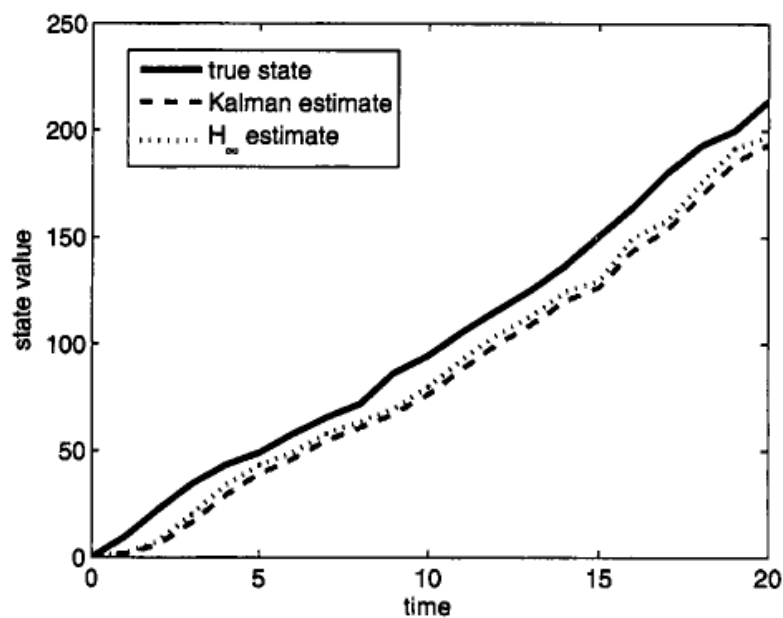


Figure 11.3 Example 11.2 results. Kalman and H_∞ filter performance when the process noise is biased. The Kalman gain is 0.62 and the H_∞ gain is 0.82. The H_∞ filter performs about 23% better than the Kalman filter.

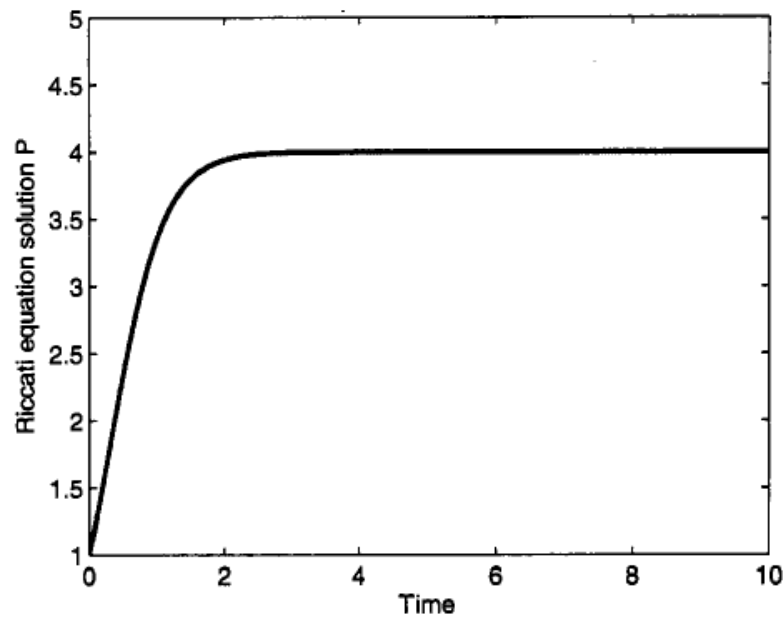


Figure 11.4 Example 11.4 H_∞ Riccati equation solution as a function of time.

time-varying Riccati solution has converged to steady state (note that the time-varying filter performs better during the initial transient). But after the Riccati solution gets close to steady state (after about $t = 1$) the performance of the two filters is nearly identical. This illustrates the possibility of saving a lot of computational effort by using a steady-state filter while giving up only an incremental amount of performance.

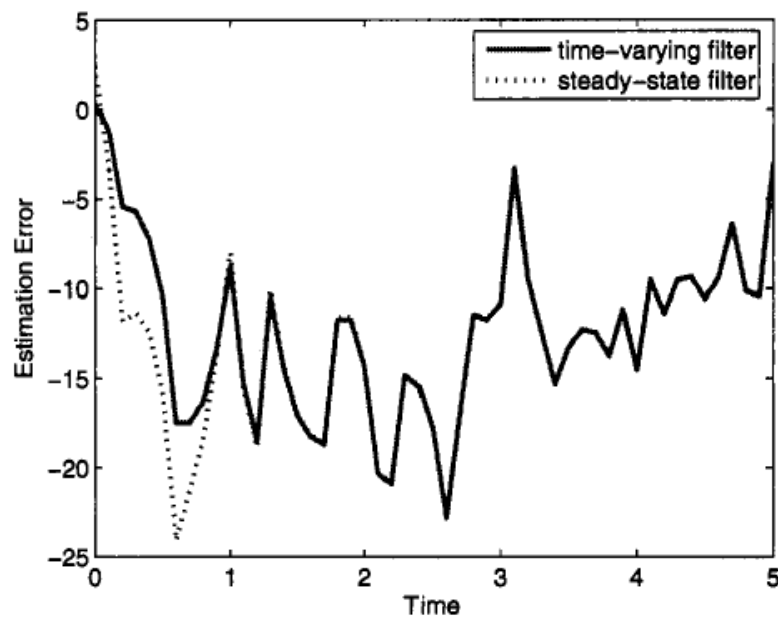


Figure 11.5 Example 11.4 time-varying and steady-state H_∞ filter performance when the measurement noise is zero-mean.

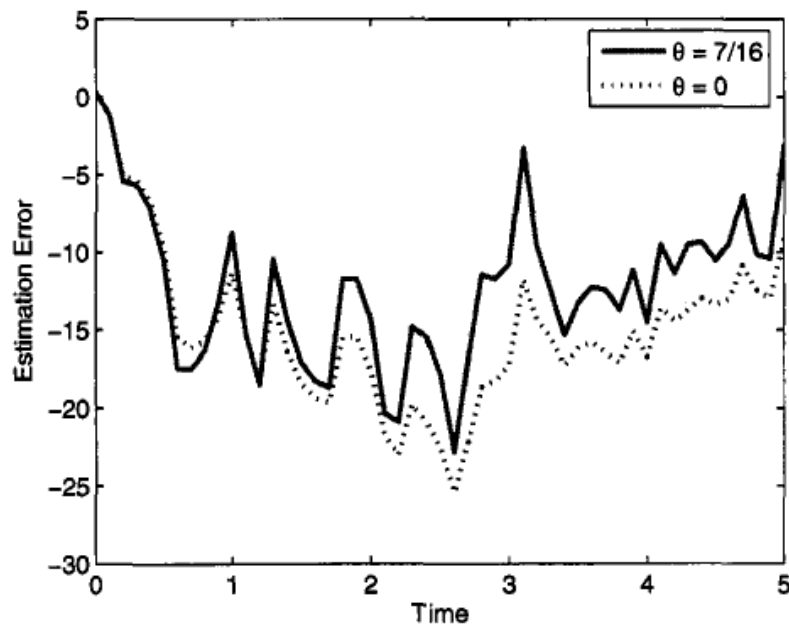


Figure 11.6 Example 11.4 time-varying Kalman and H_∞ filter performance when the measurement noise is not zero-mean.

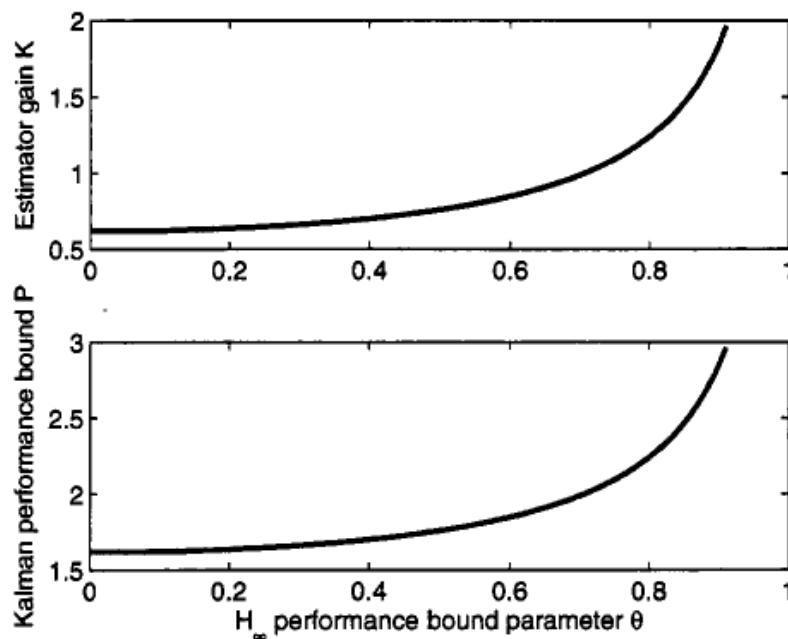


Figure 12.1 Results for Example 12.1 of the estimation-error variance bound and estimator gain as a function of θ for the mixed Kalman/ H_∞ filter. As θ increases, the worst-case performance bound $1/\theta$ decreases, the error-variance bound increases, and the estimator gain increases greater than the Kalman gain ($\theta = 0$). This shows a trade-off between worst-case performance and RMS performance.

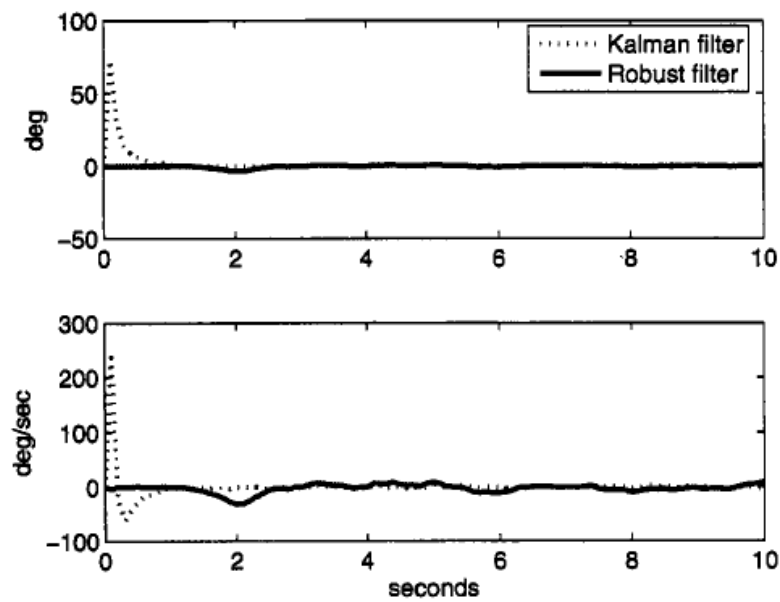


Figure 12.2 Position and velocity estimation errors for Example 12.2 for the Kalman filter and the robust filter, assuming that the system model is perfectly known. The robust filter performs better at the beginning of the simulation, but the Kalman filter performs better in steady state. The steady-state Kalman filter estimation errors have standard deviations of 0.33 deg and 1.65 deg/s, whereas the robust filter has standard deviations of 0.36 deg and 4.83 deg/s.

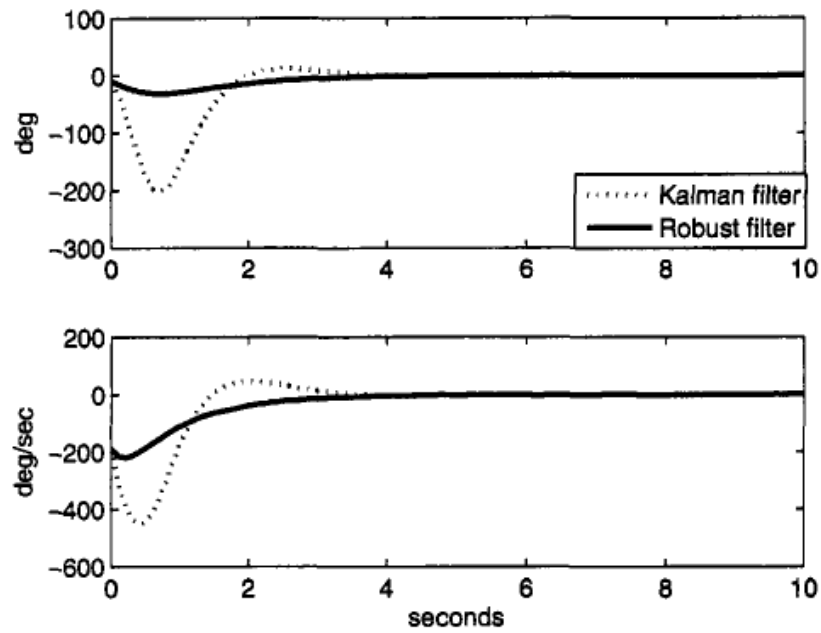


Figure 12.3 Position and velocity estimation errors for Example 12.2 for the Kalman filter and the robust filter, assuming that the system model is not well known. The robust filter performs better both at the beginning of the simulation and in steady state. The steady-state Kalman filter estimation errors have standard deviations of 0.36 deg and 1.33 deg/s, whereas the robust filter has standard deviations of 0.28 deg and 1.29 deg/s.

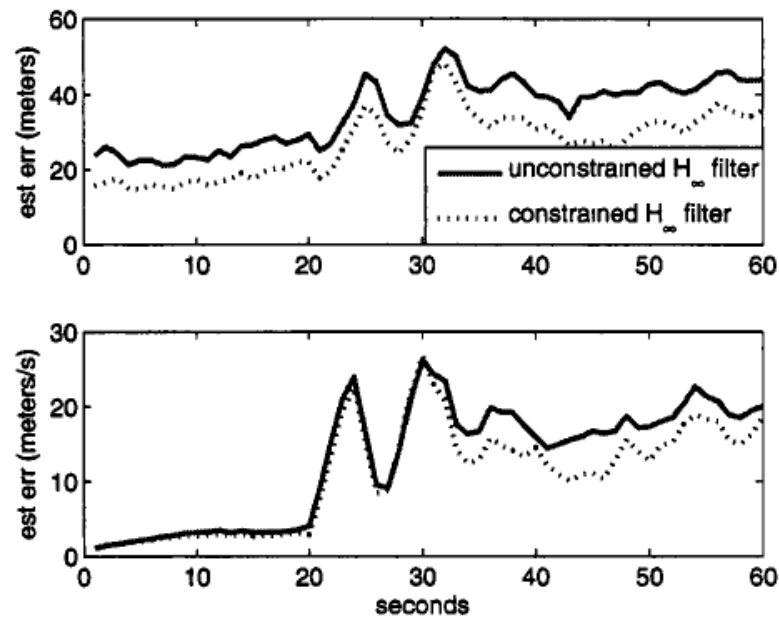


Figure 12.4 Example 12.3 unconstrained and constrained H_∞ filter estimation-error magnitudes. The plots show the average estimation-error magnitudes of 100 Monte Carlo simulations when the noise statistics are nominal.

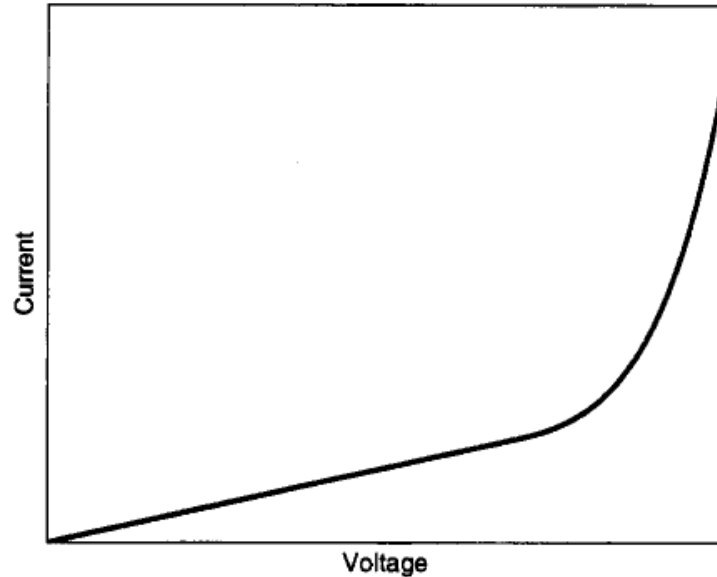


Figure 13.1 Typical current/voltage relationship for a resistor. The relationship is linear for a limited range of operation, but becomes highly nonlinear beyond that range.

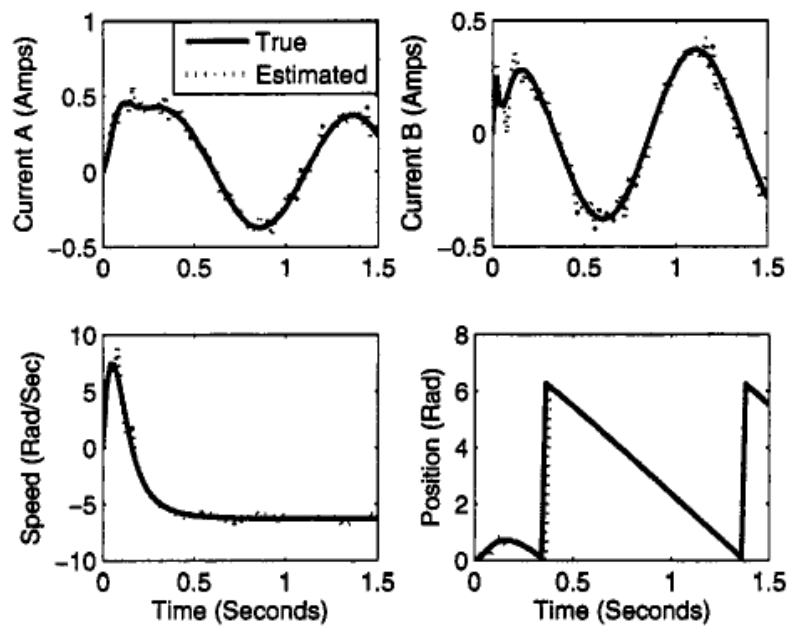


Figure 13.2 Continuous extended Kalman filter simulation results for the two-phase permanent magnet synchronous motor of Example 13.1.

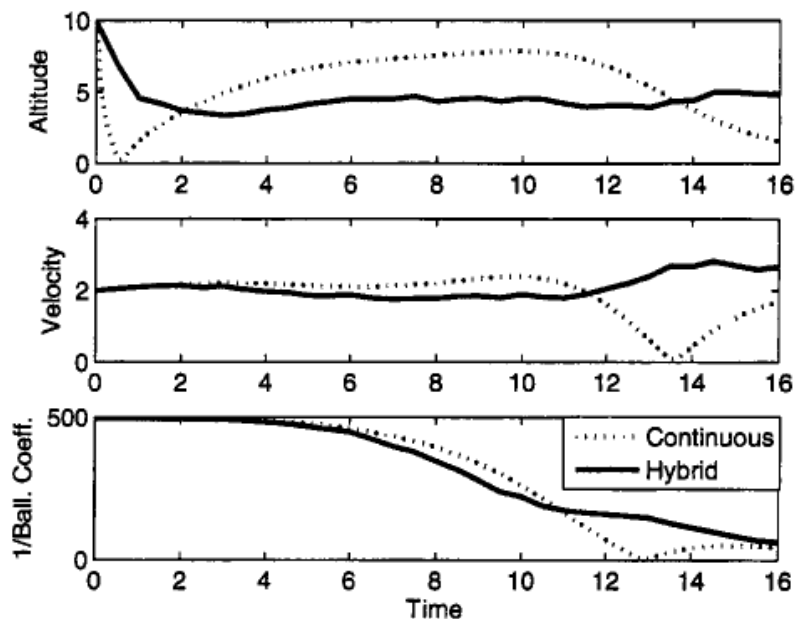


Figure 13.3 Example 13.2 altitude, velocity, and ballistic coefficient reciprocal estimation-error magnitudes of a falling body averaged over 100 simulations. The continuous-time EKF generally performs better than the hybrid EKF.

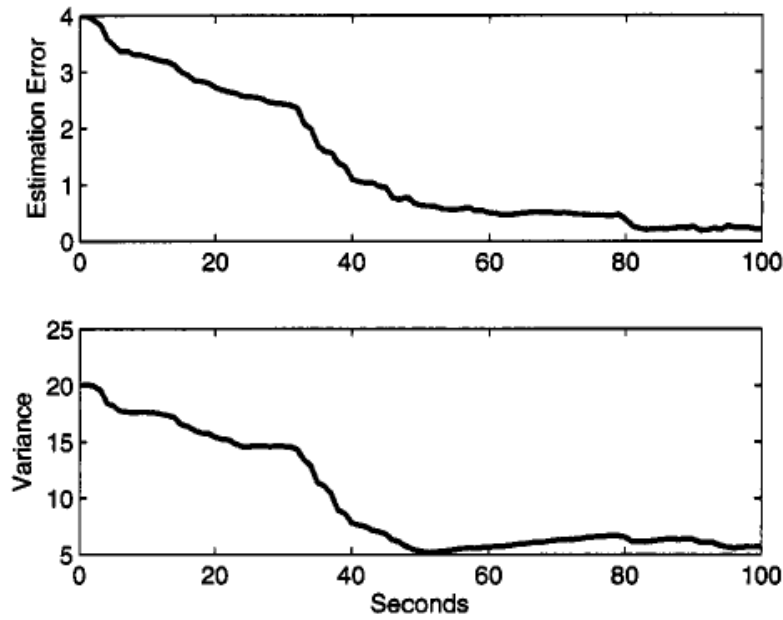


Figure 13.4 Example 13.4 results. Typical parameter estimation performance and parameter uncertainty for an extended Kalman filter estimating $-\omega_n^2$ for a second-order system. The estimation error of the unknown parameter and its variance gradually decrease toward zero.

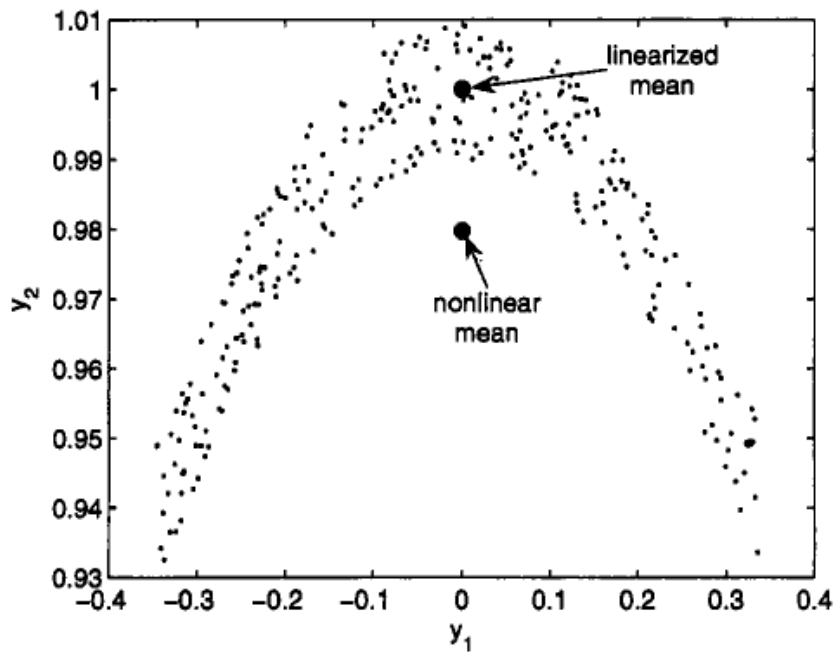


Figure 14.1 Linearized and nonlinear mean of 300 randomly generated points with $\tilde{r}$ uniformly distributed between ± 0.01 and $\tilde{\theta}$ uniformly distributed between ± 0.35 radians.

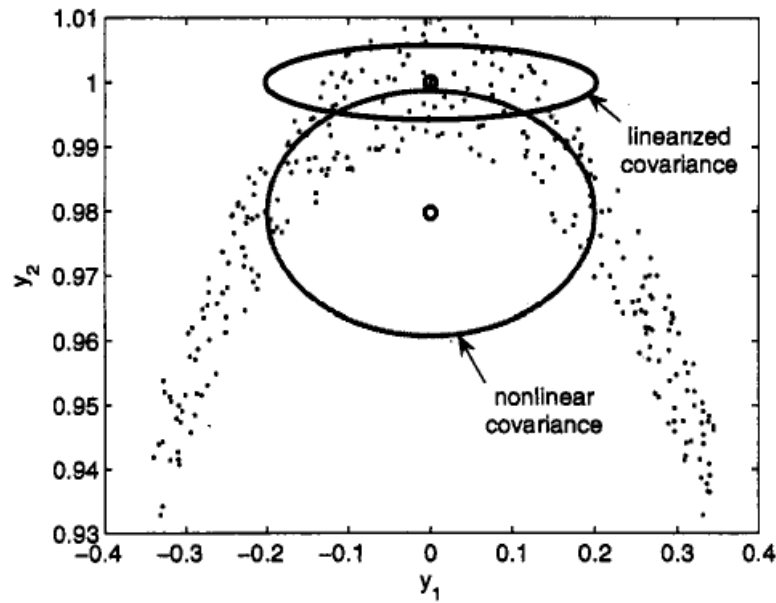


Figure 14.2 Linearized and nonlinear mean and covariance of 300 randomly generated points with $\tilde{r}$ uniformly distributed between ± 0.01 and $\tilde{\theta}$ uniformly distributed between ± 0.35 radians.

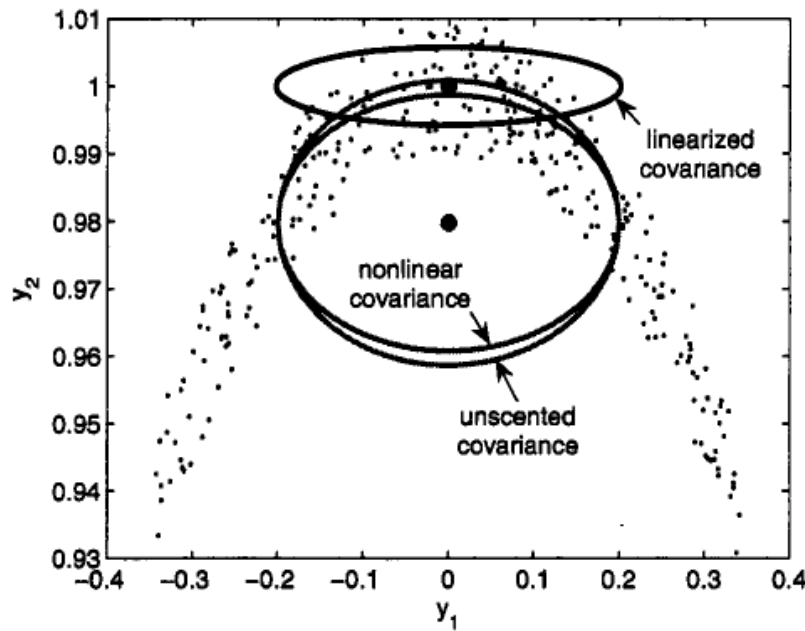


Figure 14.3 Results of Example 14.1. A comparison of the exact, linearized, and unscented mean and covariance of 300 randomly generated points with $\tilde{r}$ uniformly distributed between ± 0.01 and $\tilde{\theta}$ uniformly distributed between ± 0.35 radians.

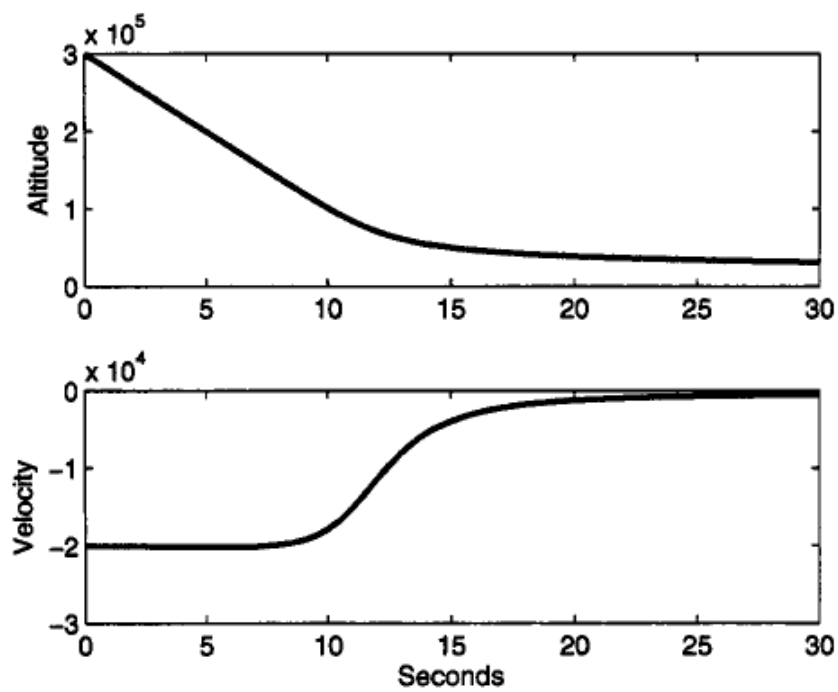


Figure 14.4 Altitude and velocity of a falling body for Example 14.2.

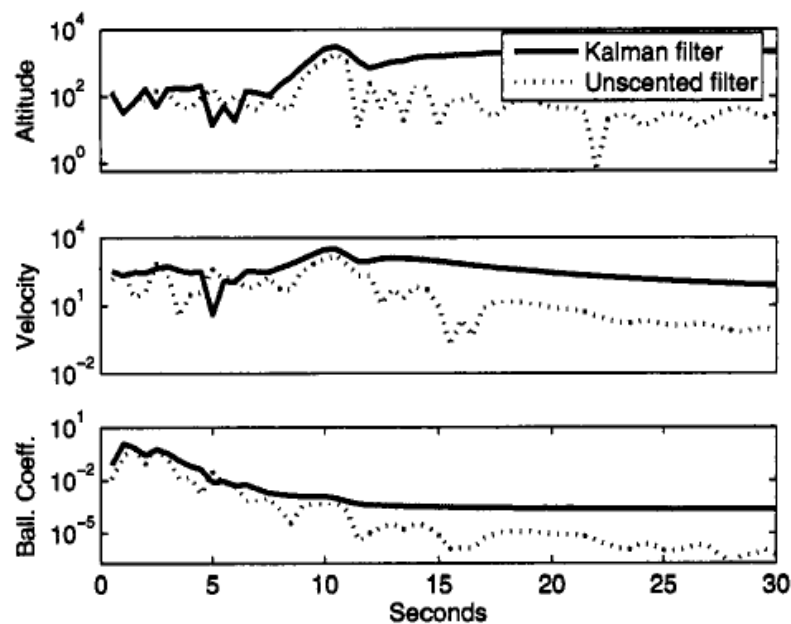


Figure 14.5 Kalman filter and unscented filter estimation-error magnitudes of the altitude, velocity, and ballistic coefficient of a falling body for Example 14.2.

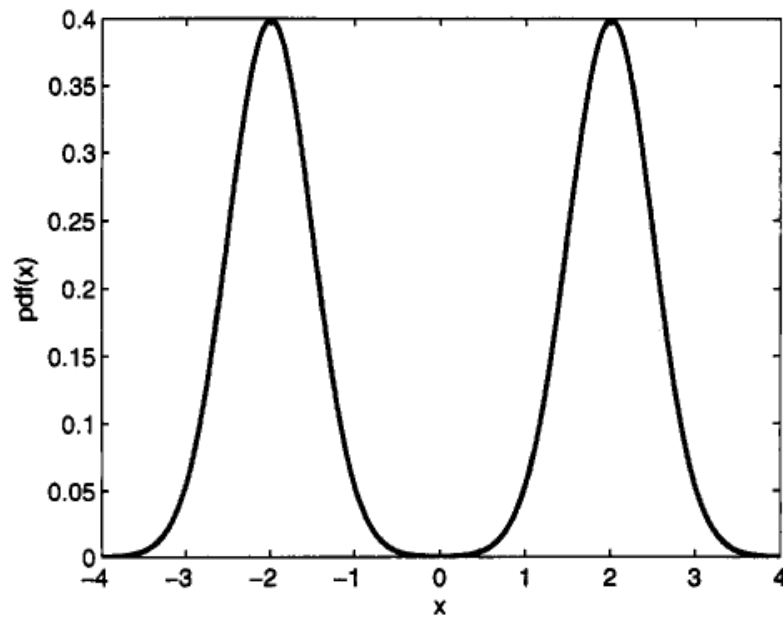


Figure 15.1 An example of a multimodal probability density function. What single number should be used as an estimate of x ?

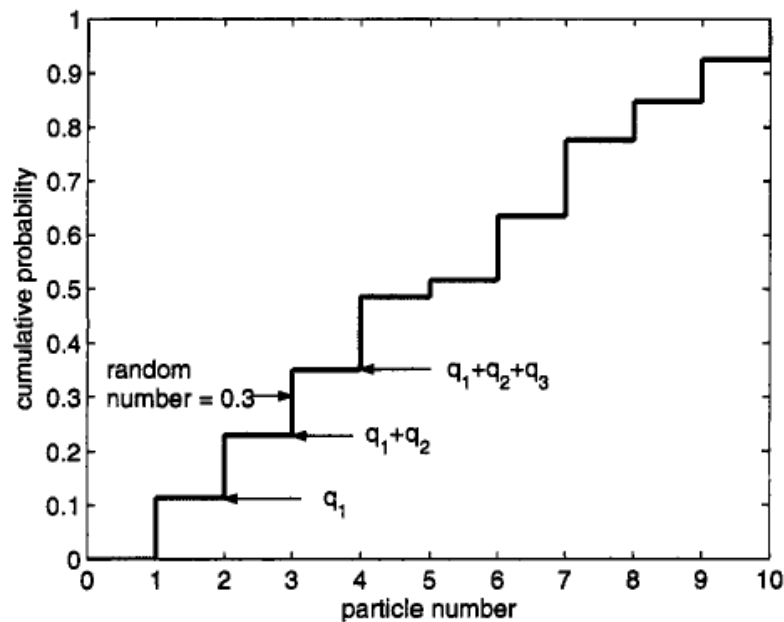


Figure 15.2 Illustration of resampling in the particle filter. For example, if a random number $r = 0.3$ is generated (from a distribution that is uniform on $[0, 1]$), the smallest value of j for which $\sum_{m=1}^j q_m \geq r$ is $j = 3$. Therefore the resampled particle is set equal to $x_{k,3}^-$.

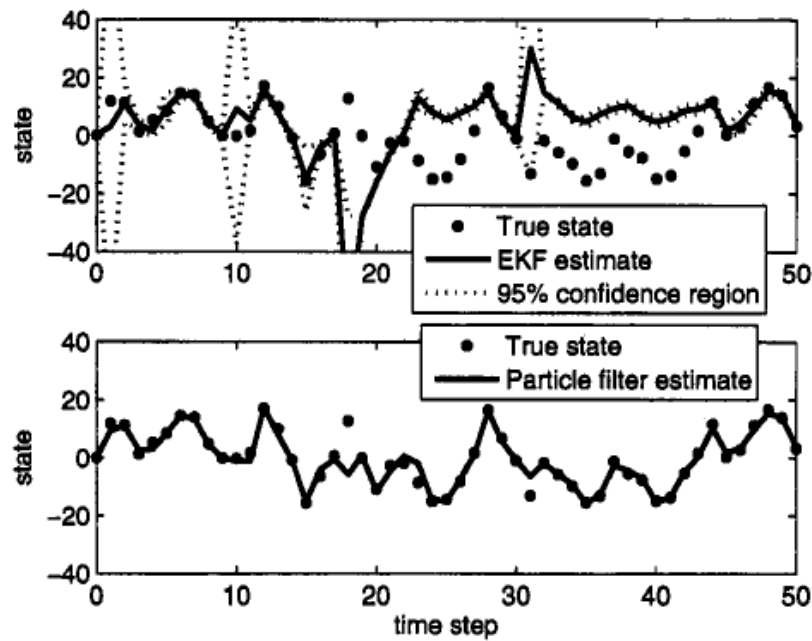


Figure 15.3 Example 15.1 results. Extended Kalman filter and particle filter estimation performance for a highly nonlinear scalar system.

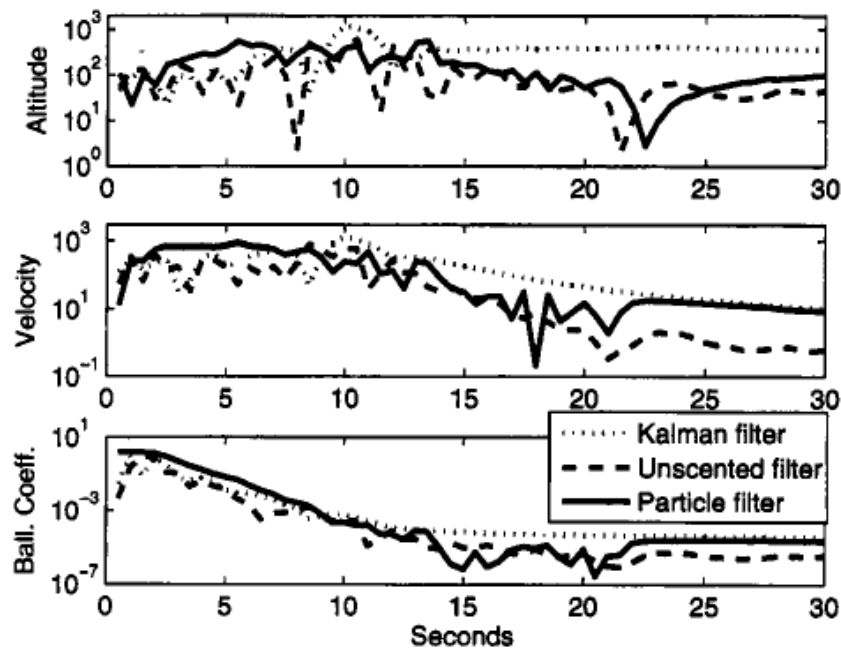


Figure 15.4 Example 15.2 results. Kalman filter, unscented filter, and particle filter estimation-error magnitudes.

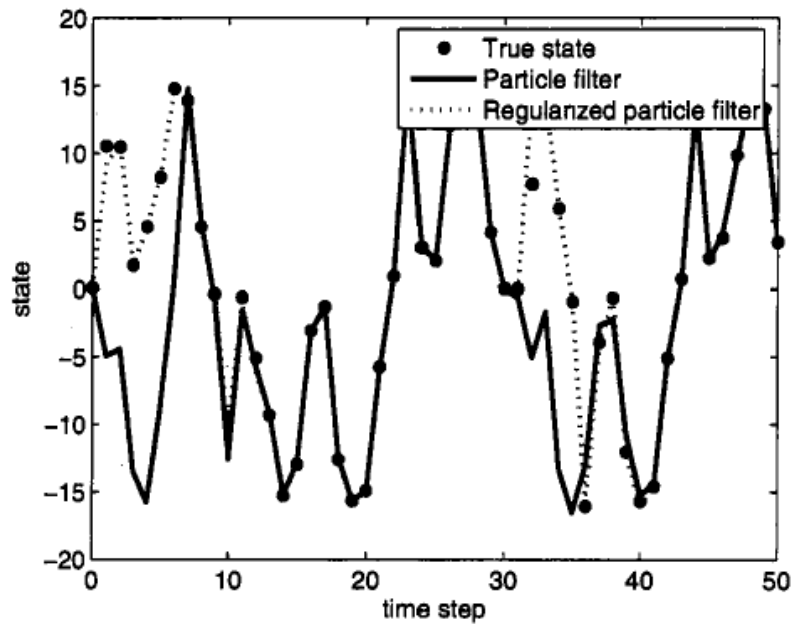


Figure 15.5 Particle filter estimation performance for the highly nonlinear scalar system of Example 15.3. This shows the improvement that is possible with the use of an RPF.

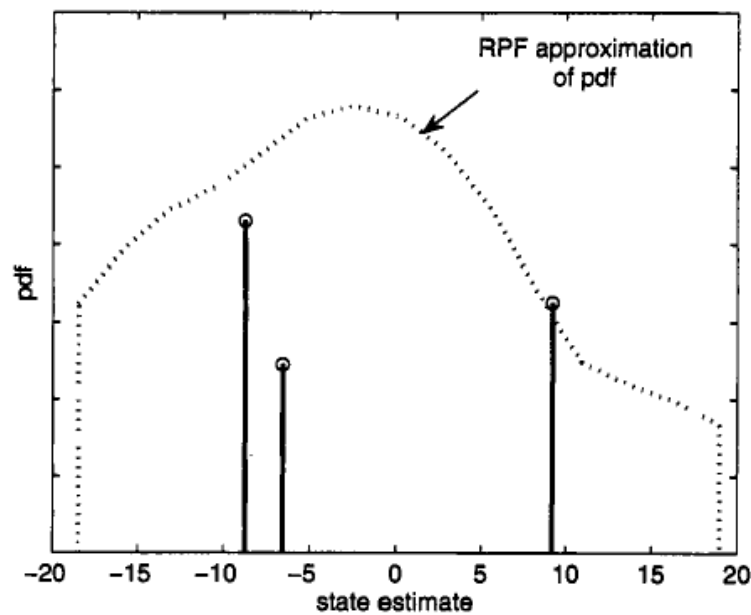
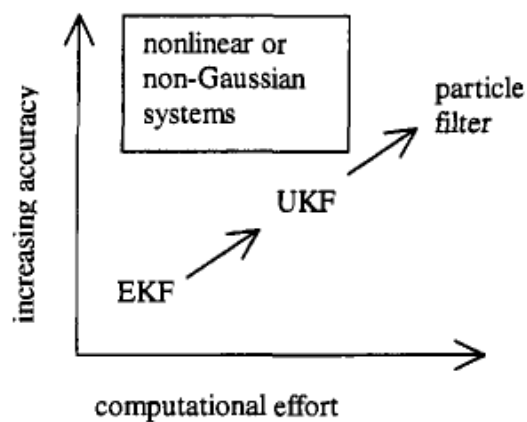
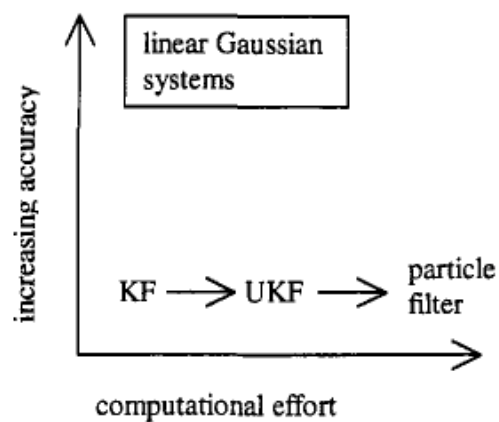


Figure 15.6 This shows the discrete pdf approximation of the standard particle filter (with three particles), and the continuous pdf approximation of the RPF. This plot is a snapshot of the pdf approximations at one time instant.



(a) The above figure depicts the increasing computational effort and increasing accuracy that is obtained by going from an EKF to a UKF to a particle filter. This applies to systems that are nonlinear or non-Gaussian.



(b) The above figure depicts the fact that the Kalman filter is optimal for linear Gaussian systems. Going from a Kalman filter to a UKF to a particle filter will increase computational effort but will not improve estimation accuracy.

Figure 15.7 State estimation trade-offs.